



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Information Technology

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch : III Semester-B. Tech Information Technology
Course Code & Title : CS3391 Object Oriented Programming
Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 5 / Layouts-Flow Pane-HBox and VBox
- **Course Outcome** : CO5
- **Topic Learning Outcome:** TLO17
- **Activity Chosen** : Learning by Teaching
- **Justification:**
 - Layouts in JavaFX is used to organize UI components on the Stage. Learning by Teaching activity is chosen to make the students to understand the various layouts in JavaFX. By this activity, Students will learn the assigned topic individually and teach the topic to their peers, which improve their communication, presentation and self-learning skills.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
 - Learning by teaching is a mode of learning where students are expected to comprehend material and then prepare presentations to teach their peers.
 - The topic JavaFX Layouts was given to two students Mr.S.Sridhar and Mr. K.Dhinesh, one week before the presentation.
 - First, the student K.Dhinesh presented the topic Flowpane layout on power point and then the student S.Sridhar explained the topic HBox and VBox layout on board presentation to their class mates.
 - The topics were delivered around 30 minutes. Finally doubts in the corresponding topic was clarified and summarized.
 - The implementation of Learning by teaching activity as shown in Figure 1 and Figure 2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO5	3	2	3	2	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	The Students will be able to understand the concept of JavaFX layouts and use the layouts in design UI applications	The Students will be able to develop GUI applications using various JavaFX layouts	The Students will gain proficiency in choosing appropriate layouts for designing interactive GUI applications	As the student is taking the class, his/her individuality can be improved and able to implement UI design for any applications individually	The Students presentation skill will be improved as they are facing the audience and teaching the concepts	The Students will be to apply the various layouts and Components in developing interactive software applications

- **Images / Screenshot of the practice:**

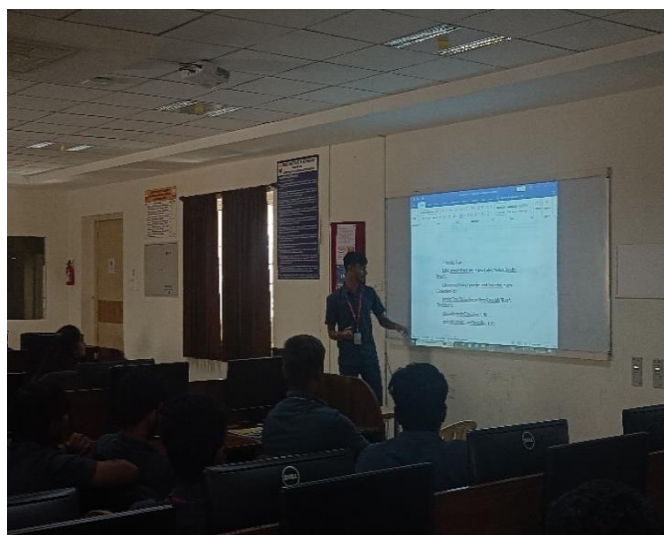


Figure 1: Screenshot of K.Dhinesh Learning by teaching activity presentation



Figure 2: Screenshot of S.Sridhar Learning by teaching activity presentation

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- All the Students were eagerly listened the class and they understand the concept of layouts in JavaFX.
- Mr.S.Sridhar and Mr.K.Dhinesh said that, the activity helps for improving their self-learning and presentation skills.
- They felt comfortable to ask doubts while teaching the topic.

- ❖ ***Benefit of the practice:***

- Learning by teaching activity improves the technical knowledge, self-learning and communication among students.
- This peer-peer learning activity is very interactive and made the slow learners to clarify the doubts with their peers.
- The Students can share their gained knowledge.

- ❖ ***Challenges faced in implementation:***

- As the activity involves only two students, unable to get all students in self-learning.
- Need to demonstrate the layout concept again to all students.
- The activity needs more time to share their views

Also, I had demonstrated the JavaFX layout with an real time example.

References:

- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1_VA_CS8603_Learning_teaching.pdf
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/6_GS_CS8592_learning_by_teaching.pdf

Signature of Faculty Member

HOD